Exp No:2 SEMMA, CRISP-DM and KDD Frameworks for Social

SEMMA

Algorithm

1. Sample: Select representative data.

2. Explore: Analyze data using statistics and visualization.

3. Modify: Transform and prepare data for modeling.

4. Model: Build a machine learning model.

5. Assess: Evaluate model performance

Code

```
# S - Sample
sample_df = df.sample(frac=0.8, random_state=42)
```

```
#explore
print(sample_df.isnull().sum())
print(sample_df.duplicated().sum())
```

```
Post_ID      0
Platform     1
Likes        1
Comments     0
Shares       0
Views        0
Sentiment    1
dtype: int64
1
```

```
#modify
sample_df = sample_df.drop_duplicates()
sample_df = sample_df.dropna()

from sklearn.preprocessing import LabelEncoder
...
```

```
Ellipsis
```

```
#model
model = LogisticRegression()
model.fit(X_train, y_train)
```

```
C:\Users\SHREE HARINI\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:473: ConvergenceW
arning: lbfgs failed to converge after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

LogisticRegression

▶ Parameters

```
#access
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.375
```

CRISP-DM

Algorithm

1. Business Understanding

2. Data Understanding

3. Data Preparation (already completed)

4. Modeling

5. Evaluation

6. Deployment

Code

```
X = df.drop("Sentiment", axis=1)
y = df["Sentiment"]

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.18181818181818182
```

KDD

Algorithm

1. Data Selection

2. Data Transformation

3. Data Mining

4. Pattern Evaluation

5. Knowledge Presentation

Code:

```
from sklearn.metrics import accuracy_score

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Knowledge discovered successfully.")
```

```
Accuracy: 0.18181818181818182
Knowledge discovered successfully.
```

Result:

The implementation of dataframeworks like semma,crisp-dm and kdd has successfully executed.